ASME 2026 - 2027 GO-KART DESIGN CHALLENGE

Problem Statement:

Design, manufacture, assemble, and validate a competitive go-kart powered by either an internal combustion engine or an electric motor. The objective is to maximize cumulative performance across five competition events: lateral acceleration, longitudinal acceleration, race track performance, design, and build quality.

Each team is responsible for developing an integrated vehicle design that balances dynamic performance, powertrain effectiveness, handling, structural integrity, manufacturability, reliability, and overall design quality. Teams must apply engineering principles, conduct appropriate analyses, evaluate design trade-offs, and validate their decisions through physical testing. The final vehicle must be functional, safe to operate, manufacturable using available resources, and completed within the established project constraints and competition schedule.

Competition Objective:

The winning go-kart will be the vehicle that achieves the highest cumulative score across the five competition events. Teams should therefore pursue a balanced design rather than optimizing for any single event at the expense of overall performance.

The competition will evaluate the following areas:

- **Lateral Acceleration**: Ability to generate lateral acceleration while navigating the designated course.
- **Longitudinal Acceleration**: Ability to accelerate effectively over the designated course.
- **Race Track Performance**: Overall vehicle performance on the designated race track.
- **Design**: Quality of the engineering design, including analysis, design justification, integration of vehicle systems, and consideration of engineering trade-offs.
- **Build Quality**: Quality of fabrication, assembly, workmanship, and implementation of the intended design.

Specific course layouts, event procedures, scoring criteria, and evaluation rubrics will be provided as they are finalized.

Design Constraints

All designs must satisfy the following constraints. Numerical limits and additional requirements will be communicated by the ASME officers as they are established.

Budget

- The total vehicle cost must not exceed the established team budget of $1400 for IC, and $2150 for EV.
- Teams must develop and maintain a bill of materials (BOM) separated by retailers documenting required components, materials, quantities, and costs.
- Design decisions should consider total cost, component availability, and the cost of replacement parts.

Vehicle Packaging and Footprint

- The completed go-kart must fit within the maximum permitted overall length of 7ft and overall width of 5ft.
- Overall dimensions must include all vehicle components and protrusions, including wheels, steering components, bumpers, and powertrain components.
- The design must accommodate the intended driver, with adequate space for seating, steering, pedal operation, and safe ingress and egress.
- All vehicle systems must be packaged within the available footprint without compromising safe operation, serviceability, or structural integrity.

Powertrain

- Each team must use the powertrain architecture assigned to its team: electric or internal combustion.
- The powertrain must be compatible with the selected chassis, drivetrain, braking system, and intended operating conditions.
- All components must operate within their specified electrical, mechanical, thermal, and manufacturer-rated limits.
- Any restrictions on motors, batteries, engines, controllers, fuel systems, or drivetrain components will be specified by ASME.

Materials, Manufacturing, and Resources

- Teams must design around the tools, equipment, materials, and manufacturing processes made available through ASME or use allotted budget to complete it externally.
- Designs must be feasible to fabricate and assemble within the project schedule and the team's available skill set.
- All material and component purchases must follow ASME procurement procedures.
- Designs should prioritize accessibility for inspection, maintenance, adjustment, and repair.

Safety

- The vehicle must be structurally sound and safe to operate under its intended test and competition conditions.
- Steering, braking, throttle, electrical systems, fuel systems, and other critical subsystems must function reliably.
- Rotating components, chains, sprockets, electrical connections, hot surfaces, and other relevant hazards must be appropriately guarded or controlled.
- Electric vehicles must incorporate appropriate battery protection, electrical isolation, wiring protection, and emergency shutdown provisions.
- Internal combustion vehicles must incorporate appropriate fuel containment, exhaust management, and heat protection.
- Drivers and team members must follow all required personal protective equipment (PPE), testing, fabrication, and operating procedures.
- Final vehicle safety requirements and operating restrictions will be inspected by ASME before testing and competition.

Schedule and Project Completion

- Teams must meet established design review, procurement, fabrication, and testing milestones.
- Designs must be sufficiently developed to support the required design reviews and timely procurement of components.
- The completed vehicle must be ready for inspection and competition by the established deadline.
- Teams must reserve sufficient time for troubleshooting, validation, and corrective work before competition.

Design Documentation

- Teams must maintain engineering documentation appropriate to their design, including CAD models, drawings, calculations, analyses, and design decisions.
- Teams must produce and maintain an accurate BOM.
- Design changes must be communicated to relevant team members and reflected in applicable documentation.
- Teams must present their design rationale and demonstrate how engineering decisions support competition performance, safety, and manufacturability.

Engineering Design Expectations

Teams are expected to approach the project as an integrated engineering design process. Each major design decision should be supported by engineering reasoning rather than relying solely on intuition or trial and error.

Teams should consider, where applicable:

- Vehicle dynamics, weight distribution, and center of gravity.
- Chassis geometry, structural stiffness, and strength.
- Steering geometry, suspension behavior, and tire performance.
- Powertrain performance, gearing, torque delivery, and efficiency.
- Braking performance and driver control.
- Component selection, packaging, and system integration.
- Manufacturability, assembly, maintainability, and reliability.
- Cost, mass, and the trade-offs between competing performance objectives.

Teams should use calculations, simulation, research, prototyping, and physical testing as appropriate to their designs and available resources.

Competition Success Criteria

A successful design will demonstrate the ability to translate engineering principles into a functional vehicle that performs competitively across all five events while satisfying the established constraints.

Teams should prioritize:

1. Maximizing cumulative competition performance.
2. Developing a safe, reliable, and structurally sound vehicle.
3. Supporting design decisions with engineering analysis and testing.
4. Manufacturing and assembling the vehicle to a high standard of quality.
5. Completing the project within the approved budget and schedule.